Robot Dynamics & Control

Assignment 1: Gravity Compensation

April, 29th, 2026

Important notes (please read carefully):

- Your Simscape models **must** have the **same configurations** as the figures, with the **same** parameters shown;

2R Non-Planar Robot (2R NP)

Initial Joint configurations shown in the figure: q1=q2=0°.

The robot configuration is shown in the last page.

For each of the 3 exercises in the Simulink section, you must start from the non-planar 2R robot model shown in the figure, which follows the **Denavit-Hanterberg convention**. The **base** frame is rotated along the x axis with an angle of -90° with respect to the **world** frame. The **Joint 2 Frame** is rotated -90° around the Z axis and -90° around the X axis of **Joint 1 Frame**

Link Parameters

Length of Link 1: 50 cm

Length of Link 2: 40 cm

Width/Height of the Links: 1 cm

A1.1 question (Static gravity compensation)

Mass of Link 1: 2.5 kg

Mass of Link 2: 3.0 kg

Mass of end-effector: 0 kg

Mass of all joints: 0 kg

Initial Joint configurations : qi=[q1, q2]=[60°, -30°]

Compute the **gravity compensation torques** τ (**in N ·m**) for all joints with respect to the **base frame**.

A1.2 question (Static gravity compensation with extra mass in Link 2)

Mass of Link 1: 1.5 kg

Mass of Link 2: 2.0 kg

Additional mass in Link 2: 1.5 kg

Position of the additional mass with respect to the Joint 2 frame : 2/3* **Link 2**

Mass of end-effector: 0.0 kg

Mass of all joints: 0 kg

Initial Joint configurations: qi=[q1, q2]=[45°, -45°]

Compute the **gravity compensation torques τ (in N ·m**) for all the joints with respect to the **base frame**, considering an **additional mass** placed on **Link 2** at two-thirds of its length.

A1.3 question (Static gravity compensation with extra mass in Link 1 and external mass in End-Effector)

Mass of Link 1: 3.0 kg

Mass of Link 2: 1.8 kg

Mass of all joints: 0kg

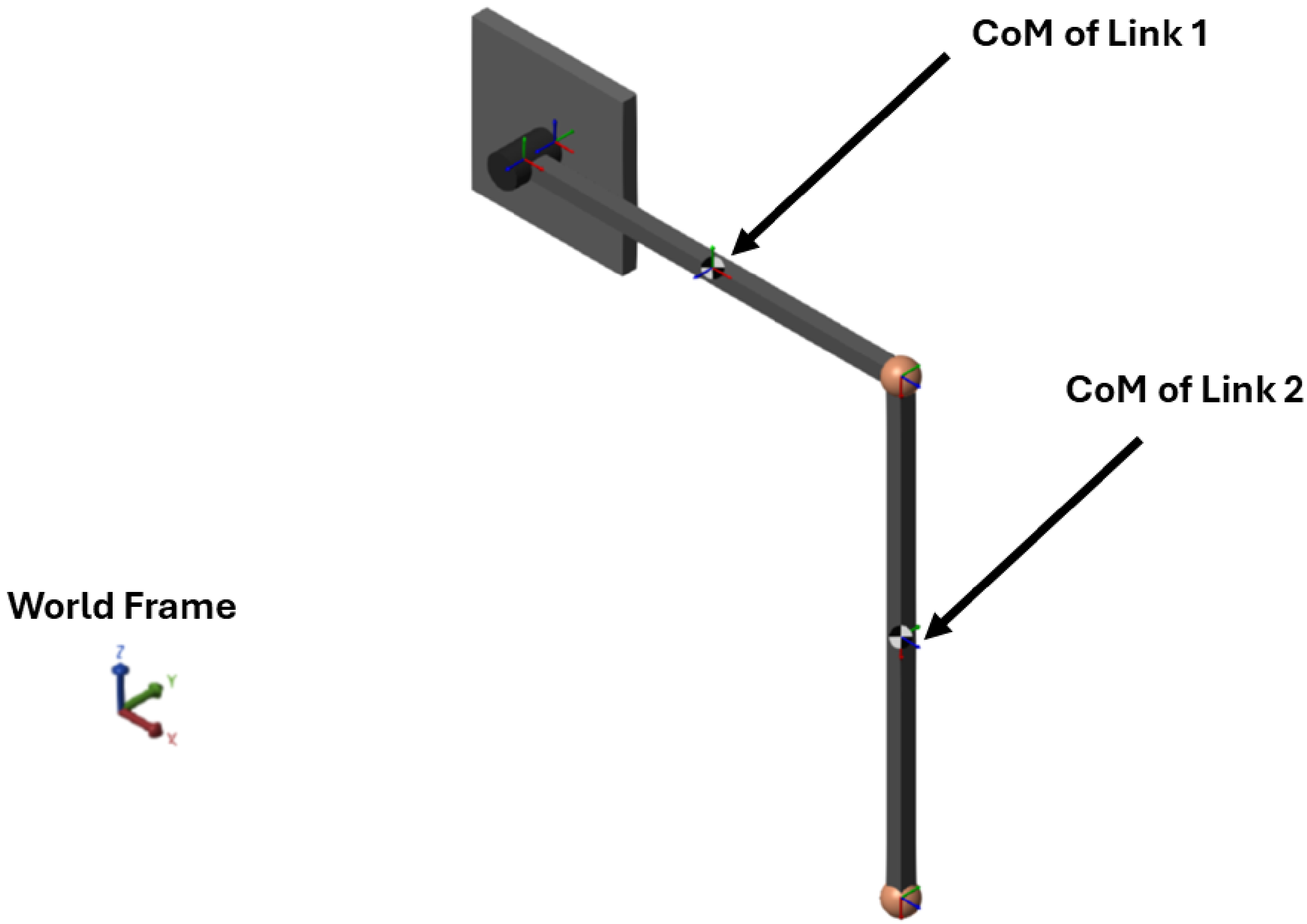
Additional mass in Link 1: 1.4 kg

Mass of end-effector: 1.2kg

Initial Joint configurations: qi=[q1, q2]=[30°,-40°]

Position of the additional mass with respect to the Joint 1 frame : 1/3* **Link 1**

Compute the **gravity compensation torques τ (in N ·m**) for all the joints with respect to the **base frame**, considering an **additional mass** placed on **Link 1** at One-thirds of its length and an external mass in the End Effector frame.


CoM of Link 1
CoM of Link 2
World Frame
Z
Y
X